

Experiment No.	6
Experiment Title	Ensemble Learning for Breast Cancer Diagnosis: Bagging, Boosting and Stacking Classifiers ¹
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1 Objective

To build and compare three ensemble learning strategies – bagging, boosting and stacking – on a binary classification task, and to identify which strategy yields the best diagnostic performance on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset.

2 Problem Statement

Given a set of 30 numeric features computed from digitized images of a fine needle aspirate (FNA) of a breast mass, the task is to classify each sample as benign or malignant. The input is a 30-dimensional feature vector per sample and the output is a binary diagnosis label (0 = Benign, 1 = Malignant). Three ensemble approaches are trained and tuned: a Bagging classifier with decision-tree base estimators, boosting classifiers (AdaBoost and Gradient Boosting, with the better of the two retained), and a stacked ensemble combining SVM, Naive Bayes and Decision Tree base learners with a logistic regression meta-learner. The goal is to evaluate and compare these models on held-out test data.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Dataset Source	<code>wdbc.data</code> , UCI Machine Learning Repository
Number of Samples	569
Number of Features	30 numeric features (the <code>id</code> column was dropped)
Number of Classes	2 (Benign = 0, Malignant = 1)
Missing Values	None found in the loaded data
Train-Test Split	80% train / 20% test (455 training samples, 114 test samples, <code>random_state=42</code>)

4 Exploratory Data Analysis

4.1 Class Distribution

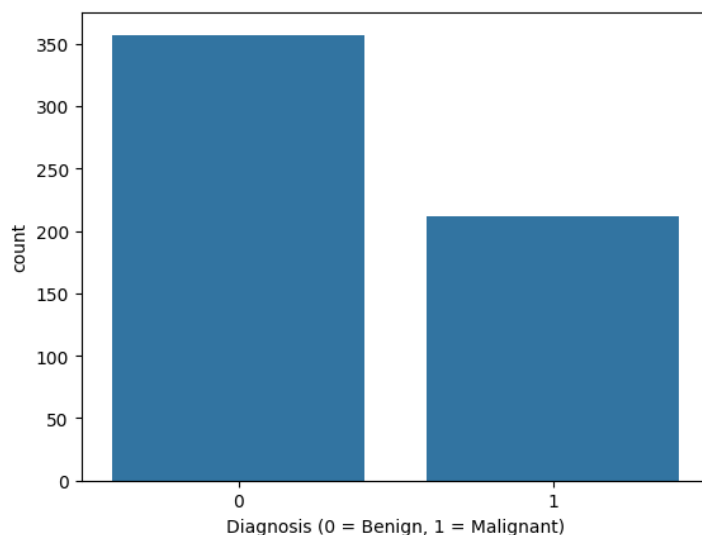


Figure 1: Distribution of diagnosis classes (0 = Benign, 1 = Malignant)

Inference:

The dataset contains 357 benign and 212 malignant samples, giving a class ratio of roughly 63:37. This is a moderate imbalance rather than a severe one, so accuracy remains a reasonably informative metric, but precision and recall are still tracked to confirm that the minority (malignant) class is not being misclassified disproportionately. The imbalance also motivates the use of stratifiable ensemble methods that can weight harder-to-classify samples, which boosting explicitly does. Overall, the class split is realistic for a screening dataset, where healthy cases outnumber diagnosed cases.

4.2 Correlation Matrix

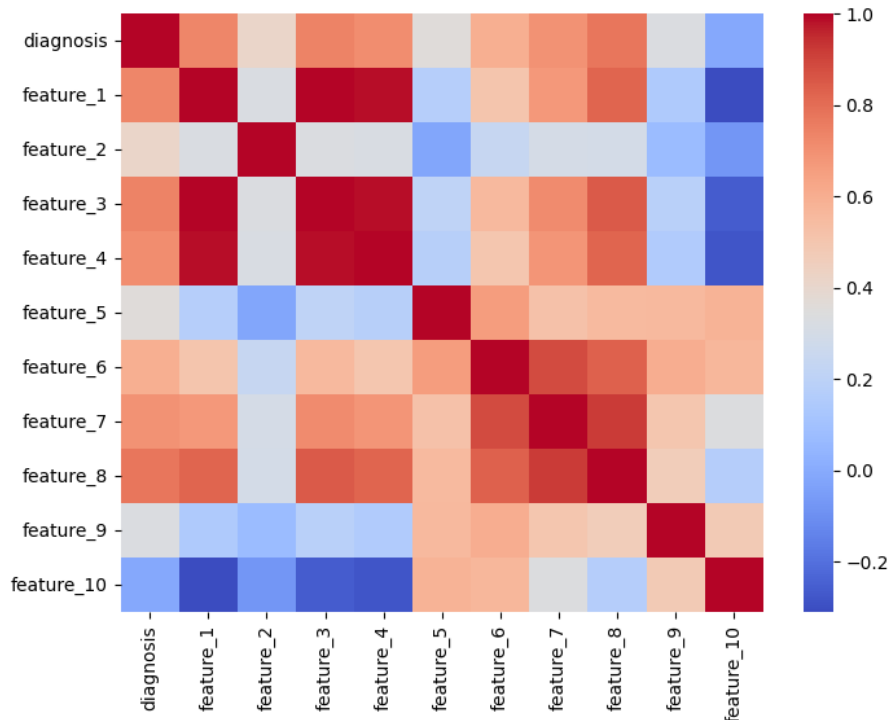


Figure 2: Correlation heatmap of the diagnosis label with the first ten features

Inference:

Several size-related features (feature_1, feature_3 and feature_4, corresponding to mean radius, perimeter and area) are strongly positively correlated with each other and with the diagnosis label, indicating that larger cell nuclei are associated with malignancy. Feature_2 (mean texture) is weakly correlated with the target, suggesting it carries less discriminative power on its own. A cluster of shape-related features (feature_6 to feature_8, e.g. compactness and concavity) also correlates positively with malignancy. This high inter-feature correlation among the size-based measurements suggests redundancy, which tree-based ensembles handle naturally through feature sub-sampling.

5 Data Preprocessing

- **Missing value handling:** The dataset was checked and contained no missing values, so no imputation was required.
- **Encoding:** The categorical target `diagnosis` was label-encoded using `LabelEncoder`, mapping Benign $\rightarrow 0$ and Malignant $\rightarrow 1$.
- **Feature scaling:** No explicit scaling (standardization/normalization) was applied. This is acceptable for the tree-based models (Bagging, AdaBoost, Gradient Boosting), which are scale-invariant, though it can affect the SVM component inside the stacked ensemble.
- **Train-test split:** The data was split 80/20 into training and test sets using `train_test_split` with `random_state=42` for reproducibility.

- **Feature engineering:** No additional feature engineering was performed; all 30 original numeric features were used directly.

6 Machine Learning Algorithm

6.1 Bagging (Bootstrap Aggregating)

Working principle: Bagging trains multiple copies of a base estimator (here, a Decision Tree) on different bootstrap samples of the training data and aggregates their predictions by majority vote. This reduces variance without significantly increasing bias.

Mathematical formulation: Given training set D of size n , B bootstrap samples $D_1, \dots, D_B$ are drawn with replacement, and models $h_1, \dots, h_B$ are trained independently. The final prediction is:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_B(x)\}$$

Advantages: Reduces overfitting and variance; parallelizable since base learners are trained independently.

Limitations: Does not reduce bias; correlated base learners limit variance reduction; less effective when the base learner is already low-variance.

6.2 Boosting (AdaBoost / Gradient Boosting)

Working principle: Boosting builds an additive ensemble sequentially, where each new weak learner focuses on the errors of the previous ensemble. AdaBoost re-weights misclassified samples; Gradient Boosting fits new learners to the residual (negative gradient) of the loss function.

Mathematical formulation: For AdaBoost, the ensemble prediction after T rounds is:

$$F(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right), \quad \alpha_t = \frac{1}{2} \ln \left(\frac{1 - \epsilon_t}{\epsilon_t} \right)$$

where ϵ_t is the weighted error of weak learner h_t .

Advantages: Reduces both bias and variance; often achieves higher accuracy than bagging on structured/tabular data.

Limitations: Sequential training prevents parallelization; more sensitive to noisy data and outliers than bagging.

6.3 Stacking

Working principle: Stacking trains several heterogeneous base learners (here, SVM, Naive Bayes and Decision Tree) and combines their predictions using a meta-learner (Logistic Regression) trained on the base learners' outputs, optionally alongside the original features (`passthrough=True`).

Mathematical formulation: If $h_1, \dots, h_k$ are base learners producing predictions $z_i = h_i(x)$, the meta-learner g produces the final output:

$$\hat{y} = g(z_1, z_2, \dots, z_k, [x])$$

Advantages: Can combine complementary strengths of different algorithm families, often outperforming any single base learner.

Limitations: Higher computational cost; risk of overfitting the meta-learner without proper cross-validation; less interpretable than a single model.

7 Experimental Setup

Python Version	3.x
Scikit-Learn Version	1.x
IDE	Jupyter Notebook
Operating System	
Hardware	

8 Hyperparameter Selection

Hyperparameters were selected using 5-fold `GridSearchCV` with accuracy as the scoring metric.

Parameter	Value
<i>Bagging (best CV accuracy = 0.9604)</i>	
n_estimators	50
max_samples	0.8
max_features	0.5
<i>AdaBoost – selected boosting model (best CV accuracy = 0.9802)</i>	
n_estimators	50
learning_rate	1
<i>Gradient Boosting (best CV accuracy = 0.9516, not selected)</i>	
n_estimators	50
learning_rate	0.1
max_depth	2
<i>Stacking</i>	
base_learners	SVM, Gaussian Naive Bayes, Decision Tree
final_estimator	Logistic Regression (max_iter=5000)
passthrough	True (CV accuracy 0.9495 vs. 0.9473 without)

9 Results

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Bagging	0.9649	0.9756	0.9302	0.9524	0.992
Boosting (AdaBoost)	0.9649	0.9756	0.9302	0.9524	0.992
Stacking	0.9561	0.9750	0.9070	0.9398	0.998

10 Result Visualization

10.1 Confusion Matrices

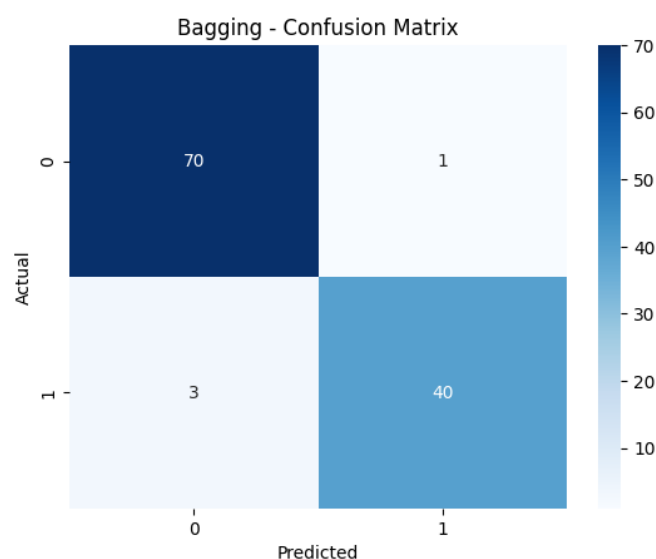


Figure 3: Confusion matrix – Bagging classifier

Inference:

The Bagging classifier correctly identifies 70 of 71 benign cases and 40 of 43 malignant cases, misclassifying only 1 benign and 3 malignant samples. The false negatives (malignant cases predicted as benign) are of particular clinical concern, but their count is low relative to the test set size. This confirms the strong CV accuracy (0.9604) generalizes well to unseen data. The single false positive shows the model rarely over-predicts malignancy.

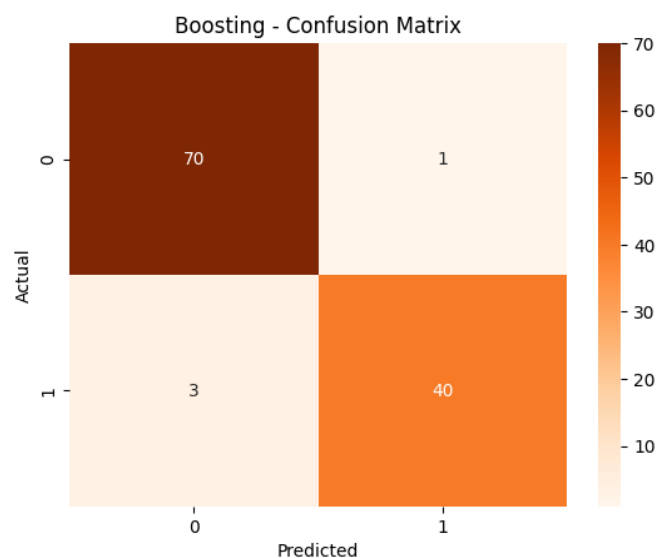


Figure 4: Confusion matrix – Boosting classifier (AdaBoost)

Inference:

AdaBoost produces an identical confusion matrix to Bagging on this test split (70/1/3/40), despite having a notably higher cross-validation accuracy (0.9802 vs. 0.9604). This suggests both models converge to similarly strong decision boundaries on this particular train-test split, even though their training dynamics differ. The consistent low false-negative count reinforces that both ensembles are reliable on the malignant class, which is the more clinically critical one.

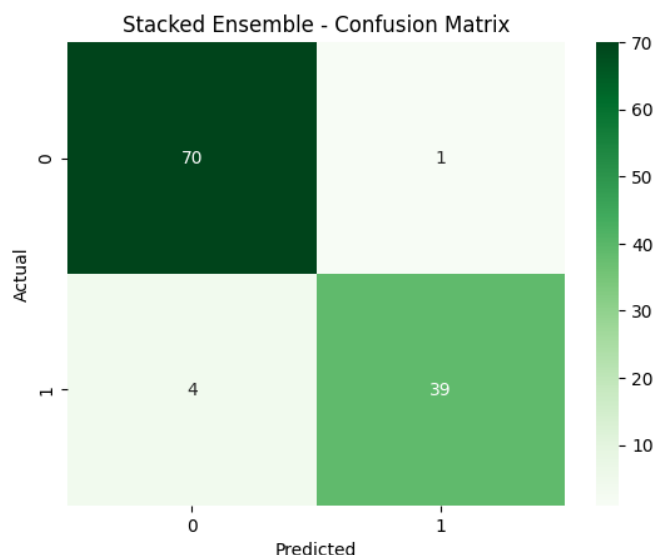


Figure 5: Confusion matrix – Stacked ensemble

Inference:

The stacked ensemble correctly classifies 70 benign and 39 malignant cases, with 1 false positive and 4 false negatives – one more false negative than Bagging and Boosting. This slightly lower recall on the malignant class explains its lower overall accuracy and F1-score in Table 2. Despite this, its very high ROC-AUC (0.998) shows the model still ranks malignant cases with high probability scores, meaning the small drop in accuracy is a threshold effect rather than a fundamental weakness in its learned representation.

10.2 ROC Curve

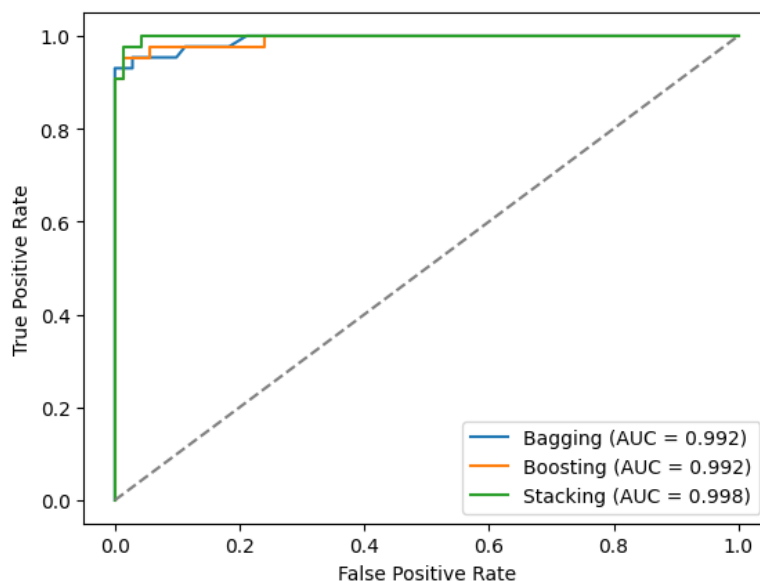


Figure 6: ROC curves for Bagging, Boosting and Stacking classifiers

Inference:

All three models achieve ROC-AUC scores above 0.99, indicating excellent separability between benign and malignant classes across all classification thresholds. The Stacked ensemble achieves the highest AUC (0.998), narrowly ahead of Bagging and Boosting (0.992 each), which is consistent with combining heterogeneous base learners to capture complementary decision boundaries. All curves rise sharply toward the top-left corner and stay well above the diagonal (random-guess) baseline, confirming that the models are strongly discriminative rather than only marginally better than chance. This is characteristic of a dataset where the classes are well separated in feature space, which matches the strong correlations observed in Figure 2.

11 Discussion

All three ensemble strategies achieved strong performance (accuracy ≥ 0.956 , ROC-AUC ≥ 0.99) on the WDBC dataset, confirming that ensemble methods are well suited to this well-separated, moderately-sized tabular problem. Bagging and the tuned AdaBoost model produced identical test-set predictions and the highest accuracy (0.9649) and F1-score (0.9524) among the three, despite AdaBoost showing a clearly higher cross-validation accuracy during tuning – a reminder that CV performance and single-split test performance can diverge on small test sets (114 samples). The stacked ensemble, despite combining three heterogeneous learners, slightly underperformed on accuracy and recall, most likely because the unscaled features disadvantage the SVM base learner, and because passthrough features add noise dimensions that a simple logistic regression meta-learner must reconcile. Its higher ROC-AUC nonetheless indicates that its underlying probability estimates are excellently calibrated for ranking, even where the default 0.5 threshold produces one extra false negative. A key limitation across all models is the

absence of feature scaling, which could particularly benefit the SVM and Logistic Regression components; adding a **StandardScaler** step and re-running the stacking grid search would likely close the small gap observed. Given the clinical context, minimizing false negatives (malignant cases missed) should be prioritized over raw accuracy in any future refinement, e.g. by adjusting the classification threshold or using recall-oriented scoring during hyperparameter search.

12 Conclusion

This experiment implemented and compared three ensemble learning paradigms – Bagging, Boosting and Stacking – for breast cancer diagnosis on the WDBC dataset. After hyperparameter tuning via 5-fold grid search, AdaBoost was selected as the strongest boosting model (CV accuracy 0.9802), and passthrough stacking outperformed non-passthrough stacking (0.9495 vs. 0.9473). On the held-out test set, Bagging and AdaBoost tied for the best accuracy (0.9649) and F1-score (0.9524), while the stacked ensemble achieved the best ROC-AUC (0.998). All three models comfortably exceeded 0.95 accuracy, demonstrating that ensemble methods – whether variance-reducing (bagging), bias-reducing (boosting), or learner-combining (stacking) – are all highly effective for this diagnostic classification task, with the choice between them depending on whether the priority is raw classification accuracy or ranking/calibration quality.

13 References

1. Sugith, “MachineLearning – Assignment 6,” GitHub repository, 2025. [Online]. Available: <https://github.com/sugith-2025/MachineLearning/tree/main/Assignment-6>